

IRE 313–Modeling and Simulation of Robotic System, Assignment-01, January 2026	
Total Marks: 30	
1.	Write a custom hardware interface class in ROS Control by inheriting from hardware_interface::RobotHW. Your class should initialize the robot, read sensor data from hardware, write actuator commands, and optionally check for controller conflicts.
2.	How can you create and integrate a custom RViz panel plugin (such as a “ButtonPanel”) using Python, a plugin XML file, and minimal CMake configuration? Also explain How to Build and Run this Plugin?
Instructions: i) Write the assignment by hand in your script/notebook. ii) Scan the completed assignment clearly. iii) Convert the scanned pages into a single PDF file. iv) Rename the PDF with your student ID and name (Example: 2201054_Mehedi.pdf) v) Submit the PDF through Google Classroom before the deadline.	